STATISTICS

What is Statistics?

Statistics is the science of **collecting**, **organizing**, **analyzing**, **interpreting**, and **presenting** data to make informed decisions.

Two Main Branches of Statistics:

1. **Descriptive Statistics** – **Summarizes** or describes characteristics of a data set.
   - E.g., average age in a class, max/min salary, standard deviation.
2. **Inferential Statistics** – Makes **predictions** or inferences about a population based on a sample.
   - E.g., predicting election results based on a sample survey.

Population & Sample :

- A **population** is the **entire group** that you want to draw conclusions about.
- A **sample** is the **specific group** that you will collect data from **population**. The size of the sample is always **less than** the total size of the population.

- It can mean a group containing elements of anything you want to study, such as objects, events, organizations, countries, species, organisms, etc.

Types Of Sampling

Technique	Description	Example
Simple Random Sampling	Every member has an equal chance of being selected.	Randomly picking 50 students from a school of 1,000 using a computer.
Stratified Sampling	Population is divided into subgroups (strata) and sampled from each.	Dividing by gender, then randomly selecting equal numbers from each group.
Systematic Sampling	Every *k*th individual is selected from a list.	Selecting every 10th name from a student roll list.
Cluster Sampling	Population is divided into clusters; some clusters are randomly selected.	Randomly selecting 3 classrooms out of 10 and surveying all students in them.

Types of data

Data is a **specific measurement of a variable**

Data is generally divided into two categories:

1. **Quantitative (Numerical):**
   - Discrete (e.g., Number of students)
   - Continuous (e.g., Height, Weight)
2. **Qualitative (Categorical):**
   - Binary (e.g., Yes/No, Head/Tail)
   - Nominal (e.g., Gender, Religion)
   - Ordinal (e.g., Rankings: Good, Average, Poor)

Type of Variable	What Does the Data Represent?	Examples
Discrete Variables (aka *integer variables*)	Counts of individual items or values.	- Number of students in a class - Number of tree species in a forest
Continuous Variables (aka *ratio variables*)	Measurements of continuous or non-finite values.	- Distance - Volume - Age

Type of Variable	What Does the Data Represent?	Examples
Binary (Dichotomous)	Yes or no outcomes (only two categories).	● Heads/Tails in a coin flip ● Win/Lose in a football game ● True/False responses
Nominal	Categories with no natural order or ranking.	Species names (Dog, Cat, Bird) Colors (Red, Green, Blue) Brands (Nike, Adidas, Puma)
Ordinal	Categories with a meaningful order, but not evenly spaced.	Finishing place in a race (1st, 2nd, 3rd) Survey ratings (Poor, Fair, Good, Excellent) Education levels (High School, Bachelor's, Master's, PhD)

Type of Variable	Definition	Best Examples
Independent Variable	The variable that is changed or controlled to test its effects.	Amount of sunlight a plant receives Type of fertilizer used Dosage of a drug
Dependent Variable	The variable that is measured; it's affected by the independent one.	Plant growth (height, number of leaves) Crop yield Patient recovery rate

- *Independent* variable = **Cause**
- *Dependent* variable = **Effect**

Missing Data, Errors, & Outliers

- **Missing data, or missing values,** occur when you don't have data stored for certain variables or participants. Data can go missing due to incomplete data entry, equipment malfunctions, lost files, and many other reasons.
- **Errors** are often inevitable, but cleansing your data helps you minimize them. If you don't remove or resolve these errors, you could end up with a false or invalid study conclusion.
- **Outliers** are extreme values that differ from most other data points in a dataset. They can have a big impact on your statistical analyses.

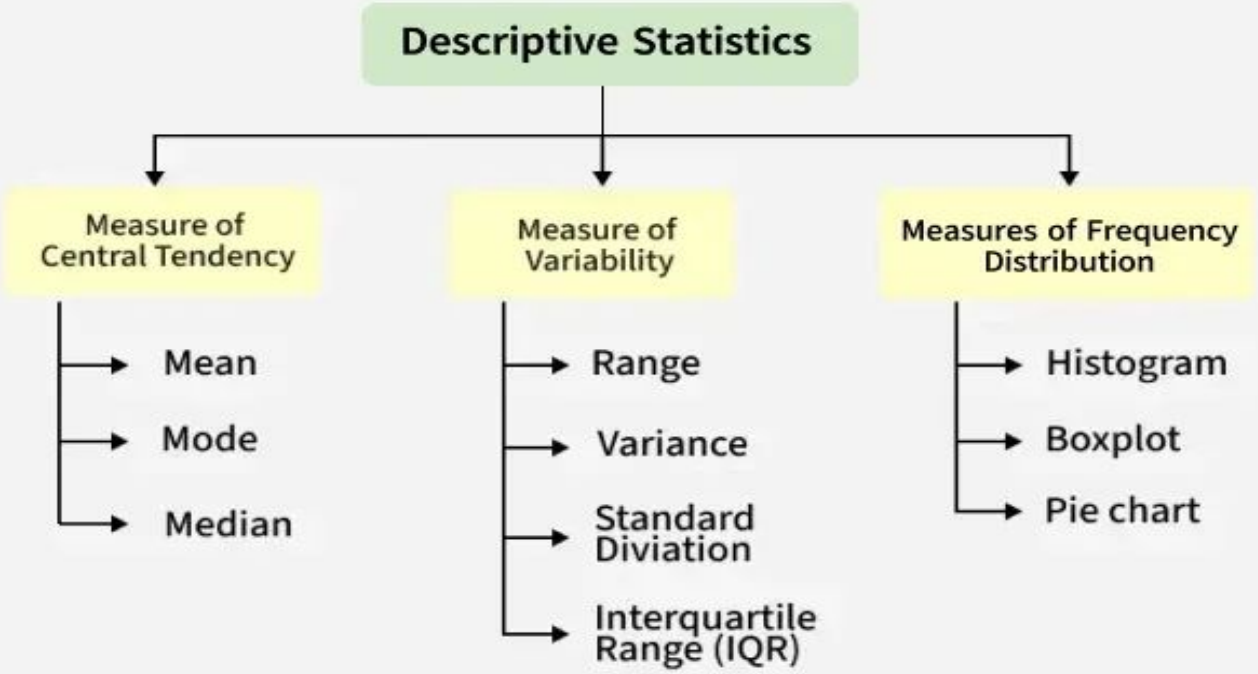
Descriptive Statistics

- **Descriptive statistics** summarize and organize characteristics of a data set. A data set is a collection of responses or observations from a sample or entire population.
- **Descriptive statistics** is a branch of statistics that **summarizes, organizes, and presents data** in a meaningful way so we can easily understand its main features.

➔ **Types of descriptive statistics**

There are 3 main types of descriptive statistics:

1. The **distribution** concerns the frequency of each value.
2. The **central tendency** concerns the averages of the values.
3. The **variability or dispersion** concerns how spread out the values are.


Descriptive Statistics
Measure of Central Tendency
Mean
Mode
Median
Measure of Variability
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Measures of Frequency Distribution
Histogram
Boxplot
Pie chart

A. Frequency Distribution

The frequency of a value is the number of times it occurs in a dataset. A **frequency distribution** is a table or chart that organizes data into categories (or intervals) and shows **how many times** each category or interval occurs.

❖ Types of Frequency Distributions

A. Ungrouped Frequency Distribution

- Used for **small datasets**.
- Lists each distinct value and its frequency.

Example (Number of books read by 10 people):

```
[2, 3, 2, 5, 3, 2, 4, 5, 3, 1]
```

Books Read	Frequency (f)
1	1
2	3
3	3
4	1
5	2

B. Grouped Frequency Distribution

- Used for **large datasets** or continuous data.
- Data is split into **class intervals**.

Steps:

1. Find **Range** = Max − Min
2. Decide **Number of Classes** (usually 5–20)
3. Find **Class Width**:
   Round up to convenient value.

   Class Width=Range / Number of Classes
4. Create intervals and count frequencies.

Test scores of 20 students:

35, 40, 42, 43, 45, 46, 48, 50, 51, 52, 55, 55, 57, 60, 62, 63, 65, 68, 70, 72

Range = 72−35=37

Choose 5 classes → Class Width = 37/5 ≈ **8**

Class Interval	Frequency (f)
35 – 42	3
43 – 50	5
51 – 58	5
59 – 66	4
67 – 74	3

C. Cumulative Frequency Distribution

- Shows **running total** of frequencies.
- Useful for finding **median** or **percentiles**.

Class Interval	f	Cumulative Frequency
35 – 42	3	3
43 – 50	5	8
51 – 58	5	13
59 – 66	4	17
67 – 74	3	20

D. Relative Frequency Distribution

- Shows proportion instead of counts.

Relative Frequency= f / N

If N = 20:

For Class 35–42 → Relative Frequency = 3/20=0.15 * 100 =(15%)

Visual Representations:

Histogram → bars showing frequency for intervals

Bar Chart → for categorical data

Pie Chart → for percentage frequencies

B. Measures of Central Tendency

These describe the **center** or **typical value** of the dataset.

1. **Mean** (Average)

$$\bar{x} = \frac{sum\ of\ \ all\ values}{number\ of\ \ all\ values}$$

2. **Median** (Middle value when data is sorted)

   For even n, take average of two middle values.

   For odd n, the middle value.

3. **Mode** (Most frequent value)

Example (Daily Study Hours for 10 students):

`[1, 2, 2, 3, 3, 3, 4, 5, 6, 8]`

Descriptive statistics can tell us:

- **Average study hours** (Mean) → `3.7` hours
- **Most common hours** (Mode) → `3` hours
- **Middle value** (Median) → `3` hours
- **Spread** (Range) → `8 - 1 = 7` hours

C. Measures of Dispersion (Variability)

These show **how spread out** the data is.

1. **Range**

The range gives you an idea of how far apart the most extreme response scores are.

Range = Max value − Min value

2. **Variance** (Average squared deviation from mean)

Formula (Population)

$$\sigma^2 = \frac{\sum (xi - \bar{x})^2}{N}$$

3. **Standard Deviation** (SD) : Square root of variance.

Smaller SD → data tightly clustered around mean.

Larger SD → data widely spread.

4. **Interquartile Range (IQR)**

$$IQR = Q3 - Q1$$

The **Interquartile Range (IQR)** is a **measure of statistical dispersion** that shows the range in which the **middle 50%** of the data lies.

Q_1 (First Quartile) = 25th percentile (value below which 25% of data lies)

Q_3 (Third Quartile) = 75th percentile (value below which 75% of data lies)

- **Range** can be misleading if extreme values are present.
- IQR focuses on the **middle bulk** of data, making it a **better spread measure** in skewed datasets.
- Helps detect outliers using the **1.5 × IQR Rule**.

Example dataset (sorted):

5, 7, 8, 12, 13, 14, 18, 21, 23, 27

Step 1: Find Q_1 and Q_3

- **Q_1**: Median of lower half = median of 5, 7, 8, 12, 13 → **8**
- **Q_3**: Median of upper half = median of 14, 18, 21, 23, 27 → **21**

Step 2: Calculate IQR

$IQR = Q3 - Q1 = 21 - 8 = 13$

4. Outlier Detection with IQR

The **1.5 × IQR rule**:

- **Lower Bound** = $Q1 - 1.5 \times IQR$
- **Upper Bound** = $Q3 + 1.5 \times IQR$

Any value **below the lower bound** or **above the upper bound** is an outlier.

Example:

- Lower Bound = $8 - 1.5(13) = 8 - 19.5 = -11.5$
- Upper Bound = $21 + 1.5(13) = 21 + 19.5 = 40.5$

  No values < -11.5 or > 40.5 → no outliers here.

Inferential Statistics

Inferential statistics is the branch of statistics that uses **data from a sample** to make **generalizations, predictions, or decisions** about a larger population.

Why Do We Use Inferential Statistics

- We **cannot measure every individual** in the population (too costly, time-consuming, or impossible).
- Instead, we collect a **sample** and **infer** results for the whole.

➢ **Types of inferential statistics:**

1) Estimation
2) Hypothesis Testing,
3) Regression Analysis
4) Correlation

Types of Inferential Statistics

1. Estimation

Used to **estimate unknown population parameters** based on sample data.

A. Point Estimation

- Provides a **single value** as the best guess of the parameter.

❖ Examples of Point Estimators

- **Sample Mean(x)**: Used to estimate the population mean (μ\muμ).
- **Sample Variance**: Used to estimate the population variance

$$s^2 = \sum_{i=1}^{n} \frac{(Xi - \bar{X})^2}{n-1}$$

- Example:
  From a sample mean of 68 kg, we estimate that the **population mean weight** is about 68 kg.

B. Interval Estimation (Confidence Intervals)

- Provides a **range of values** likely to contain the population parameter, with a certain level of confidence (e.g., 95% CI).
- Example:

  Average exam score = 75 ± 3 (95% CI → [72, 78])

  → We are 95% confident that the population mean lies between 72 and 78.

2. Hypothesis Testing

- Used to **make decisions** or **test claims** about a population.
- Hypothesis testing is a **statistical method** used to decide whether there is enough evidence in a **sample** to support or reject a claim (**hypothesis**) about a **population**.

- **Null Hypothesis (H_0) :** Statement of no effect or no difference.
- **Alternative Hypothesis (H_1) :** Statement indicating the presence of an effect or difference.

We use evidence (sample data) to decide whether to reject H_0.

Term	Meaning
Population	The entire group being studied
Sample	Subset of the population
H_0 (Null Hypothesis)	Default assumption — no difference/effect
H_1 (Alternative Hypothesis)	Opposite of H_0 — there is a difference/effect
Test Statistic	A number calculated from sample data to compare with a theoretical distribution

Term	Meaning
Significance Level (α)	Probability of rejecting H_0 when it's true (common: 0.05)
p-value	Probability of getting the observed result (or more extreme) if H_0 is true
Type I Error	Rejecting H_0 when it's true (false positive)
Type II Error	Failing to reject H_0 when it's false (false negative)

Steps in Hypothesis Testing

1. **State the hypotheses**
   - Example:
     H_0: $\mu = 50$ (mean score is 50)
     H_1: $\mu \neq 50$ (mean score is not 50)
2. **Choose significance level (α)**
   - Usually 0.05 (5% risk of being wrong).
3. **Select the right test**
   - z-test, t-test, Chi-square test, ANOVA, etc.
4. **Calculate the test statistic**
   - Based on sample data.

Find the p-value

- Compare it with α.

Make a decision

- If p-value $\leq \alpha \rightarrow$ Reject H_0
- If p-value $> \alpha \rightarrow$ Fail to reject H_0

Errors in Hypothesis Testing

When we test a hypothesis, our decision is based on sample data. Because samples are imperfect, **errors can happen**.

There are **two types of errors**:

Type I Error (α error)

Type I Error occurs when we reject a true null hypothesis.

In simple words :*We think something is happening, but in reality nothing is.*

Probability

- Probability of Type I Error = **α (significance level)**
- Common value of α = **0.05 (5%)**

Errors in Hypothesis Testing

Example:

Scenario: Exam Result Check

- **H_0 (Null Hypothesis):** The student has passed the exam.

- **H_1 (Alternative Hypothesis):** The student has failed the exam.

Type I Error:

We **declare the student failed**, but actually the student **passed**.

We rejected H_0 when H_0 was true.

Errors in Hypothesis Testing

Type II Error (β error)

Type II Error occurs when we fail to reject a false null hypothesis.

In simple words: *We think nothing is happening, but actually something is.*

Probability

- Probability of Type II Error = **β**
- Power of a test = **1 − β**

Errors in Hypothesis Testing

Example:

Same exam scenario

- **H_0:** The student has passed the exam.

- **H_1:** The student has failed the exam.

Type II Error:

We **declare the student passed**, but actually the student **failed**.

We failed to reject H_0 when H_0 was false.

Types of Hypothesis Tests

A. Based on Data Type:

- **z-test** → Large sample, known σ(Population variance)
- **t-test** → Small sample, unknown σ
  - One-sample t-test
  - Independent two-sample t-test
  - Paired t-test
- **Chi-square test** → Categorical data : tests frequency data against expectations
  - Goodness of fit
  - Independence test
- **ANOVA** → Compare means of 3+ groups

B. Based on Hypothesis Direction:

1. **Two-tailed test** → Checks for difference in both directions (≠)
2. **One-tailed test** → Checks only in one direction (> or <)

- **Two-tailed:** Detects any difference (greater or less).

  $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$

- **Right-tailed:** Detects an increase.

  $H_0: \mu = \mu_0$ vs $H_1: \mu > \mu_0$

- **Left-tailed:** Detects a decrease.

  $H_0: \mu = \mu_0$ vs $H_1: \mu < \mu_0$

Z-test vs t-test

Z-test (mean): Use when population standard deviation σ\sigma is known, or sample size is large with reliable σ\sigma-estimate

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

t-test (mean): Use when σ\sigma is unknown and you estimate with sample standard deviation s. Requires normality (or roughly so for moderate n)..

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}, \; df = n - 1$$

$$p = 2 \times P(Z > |z|)$$

$$ie., \; p = 2 \times (1 - |z|)$$

Probability Distributions

1. **Discrete Probability Distributions**: Outcomes that can be counted.
   - E.g., Rolling a die.
2. **Continuous Probability Distributions**: Outcomes that can take on any value within a range
   - E.g., Normal distribution (Gaussian), Exponential distribution.

Discrete Probability Distributions

Uses **Probability Mass Function (PMF)**: $P(X=x)$

Probabilities for each outcome must sum to **1**

Example:

- Rolling a die: outcomes = {1, 2, 3, 4, 5, 6}
- $P(X=3)=1/6$

Common discrete distributions:

- **Bernoulli distribution** – success/failure (like flipping a coin once)
- **Binomial distribution** – number of successes in n independent trials
- **Poisson distribution** – counts of events in a fixed time/space

Bernoulli distribution

When to use: A single experiment (or trial) with only two possible outcomes: success/failure, yes/no, 1/0.

Notation: X~Bernoulli(p)

- p = probability of success
- 1−p = probability of failure

PMF:

$$p(X = x) = \begin{cases} p & if\ \ x = 1(\,success) \\ 1 - p & if\ \ x = 0(\,failure) \end{cases}$$

Binomial Distribution

When to use: You repeat a **Bernoulli trial** n times independently and count the number of successes.

Notation: X~Binomial(n,p)

- n = number of trials
- p = probability of success per trial

PMF :

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where k = 0,1,2,.....n

Poisson Distribution

When to use: Counting the number of events happening in a fixed time/space interval, where events occur **independently** at a constant average rate.

Notation: X ~ Poisson(λ)

- λ = average number of events per interval

  PMF :

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where k = 0,1, 2, …….

Continuous Distributions

- **Probability Density Function (PDF):** Describes relative likelihood for different outcomes.
  Probability for an exact value is *always* zero — we look at intervals.
  Must satisfy:
  - f(x) ≥ 0 for all x
  - $$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$$
- Examples:
  - **Normal distribution** – the bell curve
  - **Exponential distribution** – time until an event occurs

Normal Distribution (Gaussian / Bell Curve)

When to use: Many natural phenomena and measurement errors follow this distribution.

Notation: X ~ N(μ,σ2)

- μ = mean (center)
- σ = standard deviation (spread)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

- Symmetrical around the mean.
- 68% of values lie within ±1σ, 95% within ±2σ, 99.7% within ±3σ.

Exponential Distribution

When to use: Models the **time until** an event occurs in a Poisson process.

Notation: X ~ Exponential(λ)

- λ = rate (events per unit time)

$$f(x) = \lambda e^{-\lambda x}, \; x \geq 0$$

Permutation & Combination

Permutations

Definition: The number of **arrangements** of items where **order matters**.

$$P(n, r) = \frac{n!}{(n - r)!}$$

n = total number of items

r = number of items chosen (arranged)

n! = n * n(-1) * (n-2) *.....* 1

Combinations

Definition: The number of **selections** of items where **order doesn't matter**.

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

Same nnn and rrr as permutations, but divide out the r!r!r! to remove orderings.